MILITARY SPECIFICATION
RELAYS, ARMATURE, for M75 ADAPTION KIT

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for hermetically sealed general purpose 5 types,and 2 classes of relays entitled Relays Armature which are used on the Nike Hercules warhead section.

1.2 Classification. This specification covers armature relays of the following types and classes:

Type I - With an arrangement consisting of one form A (see 6.1.4) and one form B (see 6.1.4) contact circuits rated at 3 amperes direct current (dc) resistance load, and two form A contact circuits rated at 12 amperes (amps) dc motor load at 30 volts (v) dc, and with a relay coil rated at 109 millamperes (ma) dc having a nominal resistance of 220 ohms.

Type II - With an arrangement consisting of two form A contact circuits rated at 3 amps dc resistive load, and two form A contact circuits rated at 12 amps dc motor load at 30 vdc, and with a relay coil at 109 ma dc having a nominal resistance of 220 ohms.

Type III- With arrangement consisting of one form A and one form B contact circuit rated at 3 amps dc resistive load, and two form A contact circuits rated at 12 amps dc motor load at 30 vdc, with a relay coil rated at 182 ma dc having a nominal resistance of 66 ohms.

Type IV - With an arrangement consisting of two form B contact circuits rated at 3 amps dc resistive load, and two form A contact circuits rated at 12 amps dc motor load at 30 vdc, resistance of 220 ohms.

Type V - An arrangement consisting of two form A contact circuits rated at 3 amps dc resistive load, and two form B contact circuits rated at 12 amps dc motor loaded at 30 vdc, and with a relay coil rated at 109 ma dc having a nominal resistance of 220 ohms.

Class I - For relays with "L" frame magnet, spacer and spring contact stacks and individual contact spring actuators.

Class II - For relays with "E" frame magnet, molded contact stacks, and one piece contact actuator.

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-571	- Solder, Tin (Tin-Lead and Lead Silvers).

MILITARY

MIL-L-73	- Lacquer, Lusterless.
MIL-W-583	- Wire Magnet, Electrical.
MIL-P-6889	- Primer Zinc-Chromate, for Aircraft Use.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1167	- Ammunition Data Card.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assembly, Procedure for.

DRAWINGS

ORDNANCE CORPS

8790179	- Relay No. 1.
8790180	- Relay No. 2.
8791622	- Relay No. 2.
8801686	- Relay, Armature.
8801687	- Relay, Armature.
8803996	- Relay, Armature.
8804004	- Relay, Armature.
EL-8815652	- Equipment Lists.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.1.1 Coil.

3.1.1.1 Magnet wire. The magnet wire shall conform to the requirements of MIL-W-583 and shall be Formvar or Formex covered wire, or approved substantial equals. In the case of Teflon insulated wire, which is not specifically referenced, MIL-W-583 shall apply except those portions applying specifically to type of insulation.

3.1.1.2 Insulators. The coil spool head, winding washers, and core insulator materials of the coil assembly shall be nylon, teflon, ceramic or an approved substantial equal. The coil lead-out wires and splice shall be insulated with nylon, mylar or an approved substantial equal.

3.1.1.3 Solder. Soft soldered electrical connections of the coil lead-out wires shall be made with Type SN50 or SN60 solder in accordance with QQ-S-571 plain core. A minimum (min) quantity of rosin flux shall be used. After soldering the joint shall be carefully cleaned and all traces of flux removed.

3.1.2 Contacts. Contact material for the light duty contacts shall be fine silver, silver cadmium (5815), or silver cadmium oxide. Contact material for the heavy duty contacts shall be silver cadmium oxide. Contact surfaces shall not be burnished or cleaned in any manner other than specified herein.

3.1.3 Contact Springs. Contact spring material shall be phosphor bronze. Garde A, spring tempered or beryllium copper, spring tempered, silver plated.

3.1.4 Contact actuators.

3.1.4.1 Contact spring and armature bushing (applicable to class 1). The contact spring bushing and armature bushing material shall be glass-teflon, teflon or an approved substantial equal.

3.1.4.2 Contact lifter (applicable to class 2). The contact lifter shall be cotton base phenolic sheet, machine grade or approved substantial equal. The phenolic sheet shall be baked 80 hours min at 140 centigrade (°C) minimum (min) at a pressure of 2.5 in. Hg maximum (max) prior to use in the relay.

3.1.5 Insulators.

3.1.5.1 Pile insulators (applicable to class 1). The pile-up insulator material shall be Alsimag, Fiber glass melamine (G-5), mycalex or an approved substantial equal.

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3.1.5.2 Contact stack insulator (applicable to class 2). Contact stack insulation material shall be glass reinforced polyester, or an approved substantial equal.

3.1.6 Header. The terminal header shall be of the hard glass compression type. The terminals shall be tinned and free from flux and excessive solder.

3.1.7 Case, mounting base and mounting bracket. The case shall be of tin coadted drawn steel. The mounting base and bracket shall be tin coated, cold-rolled steel.

3.1.8 Hook-up wire. The hook-up wire between contact springs and header terminals shall be solid tinned, bare annealed copper. The minimum allowable size of hook-up wires to terminals 3, 4, 5 and 6 shall be No. 20 AWG wire. The minimum allowable size of hook-up wires to terminals 1, 2, 7, 8, 9 and 10 shall be a No. 26 AWG wire. For wire size No. 21 AWG or smaller, teflon insulated wire or an approved substantial equal shall be utilized.

3.2 Assembly. The armature relays shall comply with all requirements specified on the drawing (dwg) 8790179, 8790180, 8791622, 8801686, 8801687, 8803996, 8804004, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Characteristics.

3.4.1 Relay life. The relay shall withstand 1000 operations without failure of a pair of contacts to open or to close properly, and shall comply with the requirements of 3.5.3.7.3.3.2 and 3.6 when tested as specified in 4.4.2.3 and 4.4.2.7 respectively.

3.4.2 Weight. The relay shall weigh 8 ounces, maximum.

3.5 Processes.

3.4.1 Tin coating. Relay headers, bases, and cases shall be tin coated. If a hot tin dip is utilized a minimum quantity of zinc chloride or similar flux shall be utilized. The flux shall be immediately neutralized and washed off. The ends of the contact springs that will be soldered shall be hot tin dipped using a minimum of rosin flux.

3.5.2 Cleaning. If the cleaning processes specified below are not utilized, alternate cleaning processes, approved by the contracting officer prior to award of contract, may be utilized.

3.5.2.1 Applicable to all parts except coil wire.

3.5.2.1.1 Applicable to class 1 only. All parts shall be subjected to the following cleaning process. Parts shall be either rocked or in some manner agitated during these operations. If parts are stored thereafter, they shall be protected from further contamination.

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a. Ultrasonically cleaned in trichloroethylene.
b. Ultrasonically cleaned in an approved detergent solution.
c. Ultrasonically rinsed in either tap or distilled water. If tap water is used parts shall be ultrasonically rinsed in distilled water to prevent drying of tap water on parts.
d. Ultrasonically rinsed in isopropyl alcohol.
e. Before the isopropyl alcohol has dried, the moisture shall be blown from the parts with high purity nitrogen.
f. Baked for a minimum period of 15 minutes at 260 ± 20° Fahrenheit (°F), under a pressure of 2.5 inches of mercury (Hg) absolute, max.

3.5.2.1.2 Applicable to class 2 only. All parts shall be subjected to the following cleaning process where specified. If parts are stored thereafter, they shall be protected from further contamination.

a. Clean in trichloroethylene.
   (1) Dip ten times in boiling trichloroethylene.
   (2) Dip ten times in continuously distilled trichloroethylene.
   (3) Air dry.
b. Alcohol pressure wash and blow dry with nitrogen.
c. Cleaning of contact tips with triple distilled chloroform.
   (1) Pressure flush.
   (2) Mechanical scrubbing using saturated lint free filter paper.
   (3) Pressure flush.
d. 40 hours bake at 140°C minimum, at a pressure of 2.5 in. Hg max.
e. 15 minute bake at 100°C minimum, at a pressure of 2.5 in. Hg max.
f. One hour bake at 100°C minimum, atmospheric pressure.
g. 7 hour bake at 2.5 in. Hg max, 1 minute, fill with nitrogen to one atmosphere pressure.

3.5.2.2 Applicable to coils.

3.5.2.2.1 Class I. In addition to the cleaning specified in 3.5.2.1 and after the coils have been assembled, they shall be ultrasonically cleaned in isopropyl alcohol, then the moisture shall be blown from then with high purity nitrogen. The coils shall be baked for forty hours minimum at 280 ± 40°F, while under a pressure of 2.5 in. of Hg absolute, or less or the coils shall be baked for three hours minimum at 260 ± 20°F, while under a pressure of 2.5 in. of Hg absolute or less while a dc or alternating current (ac) voltage is applied to the coil terminals to assure an average temperature of 260 ± 20°F, as determined by the resistance method.

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3.5.2.2.2 Class II. After the coils have been assembled clean per 3.5.2.1.2 sub-paragraphs b, d and e.

3.5.2.3 Protection of cleaned parts. After cleaning, steps shall be taken to protect the metal parts and coils from contamination such as dust, filings, metal chips, etc.

3.5.3 Sub-assembly assembling.

3.5.3.1 Contact spring assembly. The contacts shall be securely attached to the contact springs by means of welding or riveting. If welding is used the electrode shall touch only the outer periphery of the contact face. Suitable destructive tests shall be performed to insure that the welding or riveting operation is providing a good metallurgical bond over the entire joint surface. Testing shall be as specified in 4.4.2.15. The contact spring bushing shall be attached to the contact spring without the aid of cements or glue.

3.5.3.2 Relay sub-assembly Number (No.) 1.

3.5.3.2.1 Class 1. The relay sub-assembly No. 1 shall consist of the contact spring pile-up, armature and heel piece. After this assembly has been completed, it shall be cleaned as specified in 3.5.2.1.1, except that step a shall be omitted.

3.5.3.2.2 Class 2. Relay sub-assembly No. 1 shall consist of the molded contact stacks, one right hand and one left hand. After molding the stacks shall be cleaned per steps a, d, b and e in that order of 3.5.2.1.2.

3.5.3.3 Relay sub-assembly No. 2.

3.5.3.3.1 Class 1. The relay sub-assembly No. 1 shall consist of relay sub-assembly No. 1 and the coil.

3.5.3.3.2 Class 2. Relay sub-asssembly No. 2 shall consist of relay sub-assembly No.1 and the base assembly. Lead wire shall be soldered to the header pins using solder in accordance with QQ-S-571, rosin core. After soldering the joints shall be carefully cleaned and all traces of flux removed. "Contact stacks shall then be cemented together using epoxy cement per 3.5.3.5.2. The sub-assembly shall then be cleaned as specified in 3.5.2.1.2 subparagraphs b, e and d in that order".

3.5.3.3.3 Adjustments.

3.5.3.3.3.1 Applicable to class 1 only types I, II, IV and V only. The contact force of the light duty contacts and the normally open heavy duty contacts shall be adjusted to 40 grams min, and 100 grams min respectively with 64 ma dc ± 2 percent through the coil. The normally closed heavy duty contacts shall be adjusted to 60 grams min, with zero current through the coil. After this adjustment has been completed, the relay sub-assembly shall be cleaned as specified in 3.5.2.1.1 steps d and e.

3.5.3.3.3.2 Applicable to class 1 only type III only. All adjustments shall be made in accordance with 3.5.3.3.3.1 except that there shall be 130 ma dc ± 2 percent through the coil.

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3.5.3.4 Base assembly.

3.5.3.4.1 Class 1. The base, mounting bracket, and studs shall be assembled by a brazing operation. Rosin flux shall be used to solder the header to the base, and the hook-up wires to the header pins. The assembly shall be cleaned as specified in 3.5.2.1.1.

3.5.3.4.2 Class 2. The base,mounting bracket, and studs shall be assembled by a brazing or welding operation. The header shall be soldered to the base using solder in accordance with QQ-S-571, rosin core. The assembly shall be cleaned as specified in steps a and e of 3.5.2.1.2.

3.5.3.5 Relay sub-assembly No. 3.

3.5.3.5.1 Class 1. The relay sub-assembly shall consist of the relay sub-assembly No. 2 and the base assembly. The hook-up wires shall be soldered to the contact spring using rosin core solder. The contact end of the springs shall be protected with a cover during this soldering operation. After the relay sub-assembly No. 3 has been completed, it shall be cleaned in accordance with steps d, e, and f of 3.5.2.1.1.

3.5.3.5.2 Class 2. Relay sub-assembly No. 3 shall consist of relay sub-assembly No. 3 the magnet, the armature, the armature clip, contact lifter, shock springs and contact mounting screws. The armature magnet assembly shall consist of the finished coil, the magnet, the armature stop, and the back bracket. The armature stop shall be securely welded to the magnet. The coil shall be fastened to the magnet by epoxy cement. The epoxy cement shall be Shell Epon 828 with Triethylenetetramine as the catalyst or an approved substantial equal. The armature shall be seated on the magnet by means of the armature clip. The coil leads shall be soldered to the header pins using solder in accordance with QQ-S-571, rosin core. After soldering, the joints shall be carefully washed and all traces of flux removed. The magnet assembly shall be mechanically fastened to the base mounting bracket. The contact stack support arms shall be fastened to the magnet and the contact lifter placed on the contacts. Adjust contact stack arms and contact lifters so that contact gaps of 0.023 to 0.027 inches is obtained for the heavy duty contacts and 0.018 inches minimum is obtained for the light duty contacts and overtravel of 0.012 $\pm$ 0.002 inches is established for the heavy duty contacts and 0.006 inches min is established for the light duty contacts. After the relay sub-assembly No. 3 has been completed, it shall be cleaned in accordance with steps a, d, b, and e of 3.5.2.1.1 in that order.

3.5.3.6 Inline testing. The relay sub-assembly No. 3 shall be subjected to inline testing for the following characteristics and objectives. Test data shall be recorded and identified with the serialized relay. Metal stamping or tags shall be utilized for internal marking. When external marking is utilized the serial number may be stamped on the base assembly.

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3.5.3.6.1 Contact resistance normally closed (N.C) (applicable to classes 1 and 2 types I, III IV and V only). The terminal-to-terminal resistance of the N.C. contact circuit shall measure 0.030 ohms or less, before the sub-assembly is assembled into a complete relay, when tested as specified in 4.4.1.2. Accuracy of resistance measurement shall be ± 0.001 ohm.

3.5.3.6.2 Operate current. With a current measurement accuracy of ± 2 percent, the value of coil current at which the armature seats shall be observed, when tested as specified in 4.4.1.3.

3.5.3.6.3 Contact resistance normally open (N.O.). The terminal-to-terminal resistance of any pair of N.O. contact shall measure 0.030 ohm or less, which measurement shall be made upon the first closure of the contacts, before the sub-assembly is assembled into a complete relay, when tested as specified in 4.4.1.4. Accuracy of resistance measurement shall be ± 0.001 ohm.

3.5.3.6.4 Release current. With a current measurement accuracy of ± 2 percent the value of the coil current shall be observed, when tested as specified in 4.4.1.5.

3.5.3.6.5 Coil resistance.

3.5.3.6.5.1 Applicable to classes 1 and 2, types I, II IV, and V only. The resistance of the coil shall measure 220 ± 11 ohms when tested as specified in 4.4.1.1.

3.5.3.6.5.2 Applicable to classes 1 and 2, types III only. The resistance of the coil shall measure 66 ± 4 ohms when tested as specified in 4.4.1.1.

3.5.3.6.6 Contact force (applicable to classes 1 and 2). With the relay coil energized, with 64 ma dc ± 2 percent for types I, II, IV and V and 130 ma dc ± 2 percent for type III, the contact force shall be measured. Accuracy of the force measurements shall be ± 2 grams. As an alternate contact overtravel limits which have been correlated with force measurements shall be measured and recorded on a go-no-go basis.

3.5.3.7 Final cleaning and assembly of sub-assembly No. 3 and case. The following procedures shall be performed in sequence in an air conditioned ambient atmosphere, a separate area with the temperature controoled to 72 ± 3°F, the relative humidity controlled at 35 percent ± 5 percent, and the air filtered so that there are no more than 50,000 dust particles per cubic foot or hooded benches where mechanically and electrostatically filtered air shall be distributed under pressure to the hoods and the air temperature and relative humidity (rh) shall be 78 ± 5°F and 50 percent rh, max respectively. If the procedures specified in 3.5.3.7 are not utilized, alternate procedures, approved by the contracting officer prior to award

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may be utilized.

3.5.3.7.1 Cleaning.

3.5.3.7.1.1 Applicable to class 1 only. The relay sub-assembly No. 3 and cases shall be separately cleaned prior to assembly, in an ultrasonic cleaner using isopropyl alcohol. Not more than ten relays shall be cleaned in each quart of alcohol.

3.5.3.7.1.2 Applicable to class 2 only. The relay sub-assembly No. 3 and cases shall be separately cleaned prior to assembly in accordance with steps b and e in that order of 3.5.2.1.2.

3.5.3.7.2 Drying. The relay sub-assembly No. 3 and case shall be separately subjected to a dry nitrogen blast to remove the cleaning solvent. The nitrogen utilized shall be water-pumped or produced by evaporation of liquid nitrogen.

3.5.3.7.3 Assembly. Immediately upon completion of the drying operation, the case shall be placed upon the relay sub-assembly No. 3. For class 1 a cap shall be placed on the open end of the seal-off tubulation.

3.5.3.7.3.1 Sealing. For class 1 the case and base shall be joined by a crimped seam. For class 2 the relay frame bracket shall engage the case projections to assure a mechanical bond between relay and case. The seal shall be accomplished with Type SN 50 or SN 60 solder which shall not be depended upon to provide mechanical strength above that required for the seal. The case-to-base crimped joint shall first be soldered without the aid of the flux. Following this, rosin-cored solder or plain core solder and a minimum quantity of hydrazine monohydrobromide flux shall be applied to the seam to complete the seal.

3.5.3.7.3.2 Indenting. To assure that a tight fit will exist between the case and the internal mounting bracket the case shall be indented in four places for class 1 and rails shall be attached to the relay body for class 2.

3.5.3.7.3.3 Assembly completion (applicable to class 1).

3.5.3.7.3.3.1 Baking. With the cap removed from the seal-off tube, the relay seal-off tube shall be attached to a vacuum pump. The relay shall be baked for a minimum of 16 hours at 260 ± 20°F under a pressure of 2.5 in. Hg absolute minimum or the relay shall be baked for a minimum of three hours at 260 ± 20°F under a pressure of 2.5 in. Hg absolute minimum with a dc or ac voltage applied to the coil terminals to assure an average coil temperature of 260 ± 20°F, as determined by the resistance method. The relay shall then be cooled for a minimum of two hours at room temperature, while under a pressure of 2.5 in. Hg absolute or less.

3.5.3.7.3.3.2 Leakage. The relay assembly shall be evacuated to a pressure of 0.008 in. Hg absolute maximum which shall occur during a 1/2 minute vacuum cycle. Then the relay assembly shall be back-filled with nitrogen to a pressure of

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25 pounds per square inch (psi) absolute minumum, which shall occur during a 1/2 minute pressure cycle. This procedure shall be performed a total of ten times. While under a pressure of 0.0002 inch Hg absolute or less, the entire relay assembly, including seals, shall be tested for leaks by spraying the assembly with helium while the vacuum pumping system is connected to a helium-sensitive mass spectrometer. The leakage rate shall not exceed 1 x 10° cubic centimeters (at standard sea level pressure)/seconds at one atmosphere pressure differential.

3.5.3.7.3.3.3 Flushing and hermetic seal.

3.5.3.7.3.3.3.1 Flushing. The assembly shall be flushed with high purity nitrogen. After the flushing process, the relay assembly shall be evacuated to 0.0002 in. Hg absolute or less, and refilled to a pressure of at least one atmosphere, with water pumped, high purity nitrogen containing 10 percent helium. The nitrogen shall have a dew point not exceeding minus 80°F, and a maximum moisture content of 10 grains per 1000 cubic feet.

3.5.3.7.3.3.3.2 Hermetic seal. Immediately following the evacuation and refill procedures specified in 3.5.3.7.3.3.1; the seal-off tube shall be crimped or flattened, folded back over itself 180° twice and sealed using rosin core solder. The relay assembly, thus sealed, shall show no evidence of leaks, when tested as specified in 4.4.2.3.

3.5.3.7.3.4 Assembly completion (applicable to class 2 only). The substitute relay shall be baked per cleaning process g. At the end of the baking period, cool relays to room temperature under vacuum, break vacuum with high purity nitrogen produced by evaporation of liquid nitrogen and having a dew point of -80°F or less, then remove from oven and within 5 minutes drop rivet in seal-off hole and solder over hold using a minimum quantity of hydrazine flux. The relay assembly thus sealed, shall show no evidence of leaks beyond the specified limits of 6 cc or less in ten years (2 x 10-8 cc/sec) at standard sea level pressure at one atmosphere pressure differential when tested in accordance to 4.4.2.4

3.5.4 Exterior finish and marking.

3.5.4.1 Exterior finish. The exterior finish of the relay assemblies shall be as follows:

a. One coat of primer, in accordance with MIL-P-6889, Type I. (Applicable to class 1 only).
b. One coat of gray enamel, in accordance with MIL-L-73, Grade III or MIL-E-15090, Type II, Class 2, ASA 49 #620.
c. Glass, terminals, studs, and header shall not be painted.
d. The external studs shall be stainless steel 303 (applicable to class 1 only).

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3.5.4.2 Marking. The following information shall be applied to each relay assembly with suitable ink by rubber stamp or silk screen and cover with suitable varnish to TT-L-26 or MIL-V-173, Type 1, 2. The size of characters are at the supplier's option:

a. Part number.
b. Supplier's part number.
c. Supplier's name, or trade mark followed by a lot date code.
d. Serial number and lot identification.
e. Circuit diagram.

3.6 Electrical functions.

3.6.1 Contact resistance N.O. (applicable to classes 1 and 2). The pin-to-pin resistance of each N.O. contact circuit shall not exceed 0.030 ohm when tested as specified in 4.4.1.4.

3.6.2 Contact resistance N.O. (applicable to classes 1 and 2, types I, III, IV and V only). The pin-to-pin resistance of the N.O. contact circuit shall not exceed 0.030 ohm when tested as specified in 4.4.1.2.

3.6.3 Dielectric strength. The relay shall be capable of withstanding an ac potential of 500 ± 50 v root mean square (rms) at 60 Hertz between all header terminals leading to electrically isolated circuits, between all header terminals and case, and between all open pairs of contacts in both the unenergized and energized condition, and the leakage current shall not exceed 0.0015 amps ± 2 percent, when tested as specified in 4.4.1.6.

3.7 Environmental testing. The relay assembly shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The relay assembly shall also comply with 3.6, with operate current and with release current specified on the drawing before and after each environmental test. Testing shall be as specified in 4.4.2. Tests shall be conducted in the order specified.

3.7.1 High temperature operation. The relay assembly shall comply with the requirements specified in 3.6, and with the requirements for operate current, and release current specified on the drawing, when tested as specified in 4.4.2.1.

3.7.2 Low temperature operation. The relay assembly shall comply with the requirements specified in 3.6.1 and 3.6.2 and with the requirements for operate current and release current specified on the drawing, when tested as specified in 4.4.2.2.

3.7.3 Vibration. The relay assembly shall withstand vibrations having a frequency range of 10-500-10 Hertz (Hz) at an applied double amplitude of 0.036 inch ± 10 percent, or a constant peak acceleration of 10 g, in each of three mutually perpendicualr planes, without degradation of the relay and

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without chatter, and shall comply with the requirements for coil resistance, operate current, and release current specified on the drawing when tested as specified in 4.4.2.4.

3.7.4 Insulation resistance. The insulation resistance between header terminals leading to electrically isolated circuit, and between all header terminals and the case shall be not less than 500 ± 50 megohms, when tested as specified in 4.4.2.5.

3.7.5 Mechanical shock. The relay shall withstand 24 mechanical shocks applied in each direction along each of three mutually perpendicular axes without any pair of open contacts closing, without any pair of closed contacts opening and without degradation of product or contact chatter, and shall comply with the requirements specified in 3.5.3.6.5.1, 3.5.3.6.5.2, 3.6 and 3.7.4 and with the requirements for coil resistance, operate current and release current specified on drawing, when tested as specified in 4.4.2.7.

3.7.6 Sustained acceleration. The relay shall withstand a sustained acceleration without any pair of open contact closing, without chatter, and without degradation of the relay, when tested as specified in 4.4.2.8.

3.7.7 Operate and release times. The operate and release times of the relay shall not exceed 25 and 15 milliseconds respectively, when tested as specified in 4.4.2.9.

3.7.8 Contact overload. The light duty and heavy duty contacts shall make and break a resistance and an inductive load, respectively, 100 times min, without any pair of contacts failing to close or oper properly, and shall comply with the requirements specified in 3.5.3.6.5.1, 3.5.3.6.5.2 and 3.6, and with the requirements for coil resistance, operate current, and release current specified on the drawing, when tested as specified in 4.4.2.10.

3.7.9 Thermal cycle. The relay shall withstand a thermal cycle ranging from plus 160° ± 5°F to minus 80° ± 5°F, without degradation and shall comply with the requirements for 3.5.3.6.5.1, 3.5.3.6.5.2, 3.6.1 and 3.6.2 and with the requirements for coil resistance, operate current, and release currents specified on the drawing, when tested as specified in 4.4.2.11.

3.7.10 Low temperature and pressure. The relay shall withstand a low temperature and pressure cycle without degradation and shall comply with the requirements of 3.5.3.6.5.1, 3.5.3.6.5.2, 3.6.1 and 3.6.2 and with the requirements for coil resistance, operate current, release current specified on the drawing, when tested as specified in 4.4.2.12.

3.7.11 High pressure. The relay shall withstand a pressure of 45 inches Hg absolute without degradation and shall comply with the requirements specified in 3.5.3.6.5.1, 3.5.3.6.5.2, 3.6 and 3.7.4 and with the requirements for coil resistance, operate current, and release current specified on the drawing, when tested as specified in 4.4.2.13.

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3.7.12 High temperature storage. The relay shall withstand a temperature of + 160° + 5°F for 100 hours minimum without degradation, and shall comply with the requirements of 3.5.3.6.5.1, 3.5.3.6.5.2, 3.6 and 3.7.4 and with the requirements of 3.4.3.6.5.1, 3.5.3.6.5.2, 3.6 and 3.7.4 and with the requirements for coil resistance, operate current and release current specified on the drawing, when tested as specified in 4.4.2.14.

3.8 Soldering. All soldering except for coils (see 3.1.1.3), shall comply with the requirements of MIL-STD-1460.

3.9 Workmanship.

3.9.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.9.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.9.3 Burrs. Parts shall be free from burrs.

3.9.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.9.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.9.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 10 samples of contact assemblies and 24 consecutively produced relays of each type for first article testing to the testing facility designated in the invitation for bids or request for proposal.

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4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Test shall be conducted in order specified in Table I.

Table I
First article inspection and tests

Inspection or Test	Dest	Non Dest	Sub Assy	Assy	Number of Samples
1. Classification of characteristics (see 4.3.2.1)		X	X	X	All 24 relays
2. Inline testing (see 3.5.3.6)		X	X		All 24 relays
3. Leakage (see 3.5.3.7.3.3.2)		X		X	All 24 relays
4. Electrical functions (see 3.6)		X		X	All 24 relays
5. Hermetic seal (see 3.5.3.7.3.3.3.2)		X		X	All 24 relays
6. Vibration (see 3.7.3)	X			X	Relays 1 through 4
7. Insulation resistance (see 3.7.4)		X		X	Relays 1 through 4
8. Mechanical shock (see 3.7.5)	X			X	Relays 5 through 8
9. High temperature operation (see 3.7.1)		X		X	Relays 9 through 12
10. Relay life (see 3.4.1) (use 1/2 representative contacts)	X			X	Relays 9 through 12
11. Contact overload (see 3.7.8) (use remaining 1/2 contacts)	X			X	Relays 9 through 12
12. Dielectric strength (see 3.6.3)		X		X	Relays 9 through 12
13. Operate and release times (see 3.7.7)		X		X	Relays 13 through 16
14. Low temperature operation (see 3.7.2)		X		X	Relays 13 through 16
15. Low temperature and pressure (see 3.7.10)		X		X	Relays 13 through 16
16. Thermal cycle (see 3.7.9)	X			X	Relays 17 through 20
17. High pressure (see 3.7.11)		X		X	Relays 17 through 20
18. Sustained acceleration (see 3.7.6)	X			X	Relays 21 through 24
19. High pressure storage (see 3.7.12)		X		X	Relays 21 through 24

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Relay armature prior to sealing (see dwg 8790179, 8790180, 8791622, 8801686, 8801687, 8803996, and 8804004 as applicable).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Pieces of solder, insulation, metal chips or other foreign matter in relay contact assemblies, insulators, coils, headers and can	Visual
	102. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.9)	Visual

4.3.2.1.2 Relay armature (see dwg. 8790179, 8790180, 8791622, 8801686, 8801687, 8803996, and 8804004 as applicable).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:		
101.	Weight max	Balance
102.	Mounting dimension between center line of studs (1.187)	Gage
103.	Mounting dimension between center line of studs (.938)	Gage
104.	Mounting surface of studs to be in the same plane within .005 (for class 1) and .010 (for class 2)	Gage
105.	Stud locating dimension (.469) from centerline of 1.437 (for class 1) and 1.465 (for class 2) width	Gage
106.	Stud locating dimension (.500) from centerline of 1.656 (for class 1) and 1.670 (for class 2)	Gage
107.	Pitch diameter of mounting stud thread	Gage
108.	Major diameter of mounting stud thread min	Gage
109.	Length of thread (.375)	Gage
110.	Length of (1.656) max. (1.67 max. for class 2)	Gage
111.	Distance from top cover to underside of stud retaining washer (2.03 for class 1) and (2.06 max for class 2)	Gage
112.	Width (1.531 for class 1) and (1.58 max. for class 2)	Gage
113.	Diameter of header (1.078) max	Gage
114.	Marking incorrect	Visual
115.	Incorrect location of orientation mark	Visual
116.	Cracked header	Visual
117.	Dented enclosing can	Visual
118.	Foreign matter	Visual
119.	Objective evidence relating to material missing or improper (see 3.1)	Visual
120.	Serialization improper or missing (see 3.5.4.2)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.9)	Visual

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4.3.2.1.3 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation packing and packaging not in accordance with MIL-STD-1169	Visual
102.	Marking not in accordance with MIL-STD-1169	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.9)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test Classification
a.	Coil resistance	Major
b.	Contact resistance N.C.	Major
c.	Operate current	Major
d.	Contact resistance N.O.	Major
e.	Release current	Major
f.	Hermetic seal	Major
g.	High temperature operation	Major
h.	Low temperature operation	Major
i.	Insulation resistance	Major
j.	Dielectric strength	Major
k.	Operate and release times	Major
l.	Low temperature and pressure	Major
m.	Vibration	Major
n.	High pressure	Major
o.	Thermal cycle	Major
p.	Mechanical shock	Major
q.	High temperature storage	Major
r.	Sustained acceleration	Major
s.	Relay life	Major
t.	Contact overload	Major

Of these tests, m, p, r, s and t are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

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4.3.3.1 Coil resistance, contact resistance N.O., release current, hermetic seal, high temperature operation, low temperature operation, insulation resistance and dielectric strength. Each relay in the lot shall be tested in accordance with 4.4.1.1, 4.4.1.2, 4.4.1.3, 4.4.1.4, 4.4.1.5, 4.4.2.3, 4.4.2.1, 4.4.2.2, 4.4.2.5 and 4.4.1.6. If any item fails to comply with 3.5.3.6.5, 3.6.1, 3.6.2, 3.5.3.6.2, 3.5.3.7.3.5.2, 3.7.1, 3.7.2, 3.7.4 and 3.6.3, the item shall be rejected and removed form the lot.

4.3.3.2 Vibration, relay life, mechanical shock, sustained acceleration, operate and release times, contact overload, thermal cycle, low temperature and pressure, high pressure and high temperature storage. Twelve relays of the same class and type shall be selected at random from each lot of 200 relays and be subjected to the tests in Table IV. If any relay fails to comply with 3.7.3, 3.4.1 and 3.7.5 through 3.7.12, the lot shall be rejected until the cause of the failure is analyzed and approved corrective action taken. Then a second random sample of three relays from the lot that failed of the same class and type shall be tested and all samples must pass successfully the test sequence previously failed.

Table IV

Relays 1 through 3
- Operate and release time (see 4.4.2.9)
- Low temperature and pressure (see 4.4.2.12)
- Vibration (see 4.4.2.4)

Relays 4 through 6
- High pressure (see 4.4.2.13)
- Thermal cycle (see 4.4.2.11)
- Mechanical shock (see 4.4.2.7)

Relays 7 through 9
- High temperature storage (see 4.4.2.14)
- Sustained acceleration (see 4.4.2.8)

Relays 10 through 12
- Relay life (use 1/2 representative contacts) (see 4.4.2.6)
- Contact overload (use 1/2 representative contacts) (see 4.4.2.10)

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8815652).

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When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Electrical functions.

4.4.1.1 Coil resistance (see dwgs. 8790179, 8790180, 8791622, 8801686, 8801687, 8803996, 8804004, as applicable). The dc resistance of the coil shall be measured by any method found satisfactory to the procuring activity. This value shall be corrected to +68°F., to determine compliance with the requirement specified on the applicable drawing. The current shall be limited to 100 ± 2 ma.

4.4.1.2 Contact resistance N.C. (applicable to classes 1 and 2 types I, III, IV, and V only) (see 3.5.3.6.1 or 3.6.2 as applicable). With the contact current limited to 100 ± 2 ma, from a dc source of 5 1/2 ± 1/2 v., the terminal to terminal resistance of the N.C. contact circuit shall be measured. A Kelvin bridge can be utilized as an alternate method.

4.4.1.3 Operate current (see dwgs. 8790179, 8790180, 8791622, 8801686, 8801687, 8803996, 8804004 as applicable). After the relay coil has been de-energized, subsequent to the previous test, a miniature pilot lamp circuit of up to 28 volts dc (maximum) 100 ma., in series with each contact circuit, shall be used to monitor the contacts. Then slowly increase the relay coil current until the N.O. contacts close. The value of the coil current at which the last N.O. contact circuit closes shall be observed.

4.4.1.4 Contact resistance N.O. (see 3.5.3.6.3 or 3.6.1 as applicable).

4.4.1.4.1 Applicable to classes 1 and 2 types I, II, IV and V only. With zero volts across the contacts, the coil current shall be increased to 64 ma dc ± 2 percent. With the

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contact current limited to 100 ± 2 ma from a dc source of 5 1/2 + 1/2 v. the pin-to-pin resistance of each N.O. contact circuit shall be measured. A Kelvin bridge can be utilized as an alternate method.

4.4.1.4.2 Applicable to classes 1 and 2 type III only. The relay shall be tested as specified in 4.4.1.4.1, except that the coil current shall be increased to 130 ma dc ± 2 percent.

4.4.1.5 Release current.

4.4.1.5.1 Applicable to classes 1 and 2 type I, II, IV, and V only, (see dwgs. 8790179, 8790180, 8791622, 8801686, 8801687, 8803996 as applicable). With the relay coil still energized from the previous test 4.4.1.4, the coil current shall be increased to 109 ma dc ± 2 percent and then slowly decreased. The coil current at which the last N.O. contact opens shall be observed to determine compliance with the requirements specified on the applicable drawing.

4.4.1.5.2 Applicable to classes 1 and 2 type III only (see dwg. 8804004). The relay coil shall be tested as specified in 4.4.1.5.1, except that the coil current shall be increased to 182 ma dc ± 2 percent.

4.4.1.6 Dielectric strength (see 3.6.3).

4.4.1.6.1 Applicable to classes 1 and 2 types I, II, IV and V only. A test potential of 500 ± 50 vrms, 60 Hz, shall be applied for at least 1 second between all header terminals leading to electrically isolated circuits, between all header terminals and the case, and between all open pairs of contacts in both energized and unenergized conditions. The coil shall be energized with 64 ma dc ± 2 percent.

4.4.1.6.2 Applicable to classes 1 and 2 type III only. The relay shall be tested as specified in 4.4.1.6.1, except that the coil shall be energized with 130 ma dc ± 2 percent.

4.4.2 Environmental testing (see 3.7).

4.4.2.1 High temperature operation (see 3.7.1). The relay shall be subjected to a temperature of +160 ± 5°F. for a min. time period of 3 hours. At the end of this period, and while at this temperature, the relay shall be tested as specified in 4.4.1 except for coil resistance (4.4.1.1).

4.4.2.2 Low temperature operation (see 3.7.2). The relay shall be subjected to a temperature of -65 ± 5°F. for a min. time period of 3 hours. At the end of this period, and while at this temperature, the relay shall be tested as specified in 4.4.1 except for dielectric strength, (4.4.1.6), and coil resistance (4.4.1.1).

4.4.2.3 Hermetic seal (see 3.5.3.7.3.5.2). The sealed relay assembly shall be tested utilizing the leak detection method involving the passage of a tracer gas from one side of a presumed leak to the other. And provide a method of determining the leakage rate specified with a minimum sensitivity of instrument of 1 x 10^{-8} cubic centimeters per second or the "Radiflo" method utilizing radioactive krypton is an accepted method.

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4.4.2.4 Vibration (see 3.7.3). The relay shall be secured by its normal mounting means, to a fixture and vibrated at the specified frequency range, doubled amplitude or constant peak acceleration, and in the specified planes. Each cycle shall last 15 minutes minimum. The duration of vibration along each of the specified planes shall be 120 minutes minimum for each plane. On alternate cycles of frequency transversing, the coil shall be energized, with 64 ma dc ± 2 percent for types I, II, IV and V, and 130 ma dc ± 2 percent for type III and all relay contacts shall be monitored for chatter. On all other cycles the coil shall be de-energized and all relay contacts shall be monitored for chatter. During these tests, the contacts shall be monitored by means of a contact chatter indicator capable of detecting contact chatter of 10 microseconds or more. At the conclusion of this test, the relays shall be tested as specified in 4.4.1, 4.4.2.3 and 4.4.2.5.

4.4.2.5 Insulation resistance (see 3.7.4). The relay shall be subjected to a temperature of +100 ± 2°F., for a min time period of 3 hours. At the end of this period and while at this temperature, the insulation resistance between header terminals leading to electrically isolated circuits, and between all header terminals and the case shall be measured one at a time, using a potential of 500 ± 50 vdc. The header shall be clean before measurements are made.

4.4.2.6 Relay life (see 3.4.1).

4.4.2.6.1 Applicable to classes 1 and 2 types I, II, IV and V only. The relays shall be divided into two equal groups and these groups shall be designated as Group IIA and Group IIB. All coils shall be energized with 109 ma dc ± 2 percent. The "on" and "off" times shall be approximately 20 ± 1 second each. The contact loads shall be as specified in the following break-down. The conclusion of the test, each relay shall be tested as specified in 4.4.1 and 4.4.2.3.

4.4.2.6.1.1 Group IIA - light duty contacts.

4.4.2.6.1.1.1 Terminals 1 and 2. The contacts shall make, carry, and break a 2 ± 0.1 amps dc resistance load at 30 vdc maximum.

4.4.2.6.1.1.2 Terminals 7 and 8. Zero current.

4.4.2.6.1.2 Group IIA.

4.4.2.6.1.2.1 Heavy duty form A contacts only.

4.4.2.6.1.2.1.1 Terminals 3 and 4. The contacts shall make an inductive load at 30 vdc, maximum; having a 75 ± 2 amp surge current (1.0 millisecond rise time, maximum), that lasts for 0.3 second, minimum. At the end of the surge period, the external circuit shall switch the relay contact circuit to 12 ± .5 amps, inductive load at 30 vdc, maximum, using the same inductance as during initial surge.

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4.4.2.6.1.2.1.2 Terminals 5 and 6. Zero current.

4.4.2.6.1.2.2 Heavy duty form B contacts only.

4.4.2.6.1.2.2.1 Terminals 3 and 4. The contacts shall close with 0 volts across them. Approximately 1 second later an external circuit shall cause an inductive load at 30 vdc max having 5 ± 0.5 amp. inrush current (1.0 milliseconds rise time maximum), that lasts for 0.3 seconds minimum. At the end of the inrush surge period an external circuit shall switch the relay contact circuit to 12 ± 0.5 amps, inductive load at 30 vdc maximum using the same inductance as during inrush. After 20 seconds the relay contact shall break the inductive load. After 20 seconds the contact voltage is reduced to 10 ± 0.5 volts.

4.4.2.6.1.2.2.2 Heavy duty form B contacts only.

4.4.2.6.1.3 Group IIB - light duty contacts.

4.4.2.6.1.3.1 Terminals 1 and 2 Zero current.

4.4.2.6.1.3.2 Terminals 7 and 8. The contacts shall make, carry, and break 3 ± 0.3 amps. dc resistive load at 30 vdc maximum.

4.4.2.6.1.4 Group IIB.

4.4.2.6.1.4.1 Heavy duty form A contacts only.

4.4.2.6.1.4.1.1 Terminals 3 and 4. Zero current.

4.4.2.6.1.4.1.2 Terminals 5 and 6. These terminals shall be subjected to the procedure specified in 4.4.2.6.1.2.1.1.

4.4.2.6.1.4.2 Heavy duty form B contacts only.

4.4.2.6.1.4.2.1 Terminals 3 and 4. Zero current.

4.4.2.6.1.4.2.2 Terminals 5 and 6. These terminals shall be subjected to the procedure specified in 4.4.2.6.1.2.2.1.

4.4.2.6.2 Applicable to classes 1 and 2 type III only. The relays shall be divided, designated, and tested in accordance with 4.4.2.6.1, except that the coils shall be energized with 182 ± 12 ma dc. The contact loads shall be as specified in 4.4.2.6.1.

4.4.2.7 Mechanical shock (see 3.7.5).

4.4.2.7.1 Applicable to classes 1 and 2 types I, II, IV and V only. The relay shall be rigidly secured, by its normal mounting means, to a shock testing machine and shall be subjected to four shocks of 30 g minimum for 11 ± 1 milliseconds in each direction along the specified planes. On alternate shocks the coil shall be de-energized, and energized with 64 ma ± 2 percent, and all contacts shall be monitored by means of a chatter tester capable of detecting chatter of ten microseconds duration. After completion of the shock test, the relay shall be tested as specified in 4.4.1, 4.4.2.3 and 4.4.2.5.

4.4.2.7.2 Applicable to classes 1 and 2 type III only. The relay shall be tested as specified in 4.4.2.7.1 except that the coil shall be energized with 130 ± 10 ma dc.

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4.4.2.8 Sustained acceleration (see 3.7.6).

4.4.2.8.1 Applicable to classes 1 and 2 types I, II, IV and V only. The relay shall be rigidly secured, by its normal mounting means, to a centrifuge, in six positions successively, such that each direction along each of the three mutually perpendicular major axes, in turn extend in a radial direction with respect to the center of rotation of the centrifuge, and shall be subjected to a sustained acceleration of 50 g min for at least one minute, in each direction. During each test, the coil shall be de-energized and energized for 15 seconds min intervals with 87 ma dc ± 2 percent and all contacts shall be continuously monitored in both conditions by a chatter detector capable of detecting chatter of ten microseconds duration.

4.4.2.8.2 Applicable to classes 1 and 2 type III only. The relay shall be tested as specified in 4.4.2.8.1, except that the coil shall be energized with 146 ma dc + 2 percent.

4.4.2.9 Operate and release time (see 3.7.7).

4.4.2.9.1 Applicable to classes 1 and 2 types I, II, IV and V only. An electrical circuit shall be connected to the contacts of the relay so that the closing time, opening time and duration of the contact bounce transients and the release time of the contacts can be observed, and timed by an oscilloscope, or equivalent means. The power supply shall be regulated, and the source impedance to the relay coil shall be less than 10 ohms. The coil shall be energized with 109 ma dc ± 2 percent. The contact bounce transient, resulting from contact closure, shall be damped out within three milliseconds. The operate time shall include all contact bounce transient time.

4.4.2.9.2 Applicable to classes 1 and 2 type III only. The relays shall be tested as specified in 4.4.2.9.1, except that the coil shall be energized with 182 ma dc ± 2 percent.

4.4.2.10 Contact overload (see 3.7.8).

4.4.2.10.1 Applicable to classes 1 and 2 types I, II, IV and V only. The remaining contacts of Groups (A) and (B) of the relay life test shall be tested as follows: All coils shall be energized with 109 ma dc ± 2 percent. The "on" and "off" times shall be in approximately 20 ± 2 seconds each. The contact loads shall be in accordance with the following break down. At the conclusion of the test, each relay shall comply with the requirements specified in 4.4.1 and 4.4.2.3.

4.4.2.10.1.1 Group A - light duty contacts.

4.4.2.10.1.1.1 Terminals 7 and 8. The contacts shall make and break a 12 ± 0.5 amps, dc resistive load at 30 vdc maximum.

4.4.2.10.1.1.2 Terminals 1 and 2. Zero current.

4.4.2.10.1.2 Group A.

4.4.2.10.1.2.1 Heavy duty form A contacts only.

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4.4.2.10.1.2.1.1 Terminals 5 and 6. The contacts shall make an inductive load at 30 vdc max, having a 112.5 ± 10 amp. inrush surge current (1.0 millisecond rise time maximum) that lasts 0.3 seconds minimum. At the end of the inrush surge period, an external circuit shall switch the relay contacts to a 24 ± 1 amp. inductive load at 30 vdc maximum using the same inductance as during the inrush.

4.4.2.10.1.2.1.2 Terminals 3 and 4. Zero current.

4.4.2.10.1.2.2 Heavy duty form B contacts only.

4.4.2.10.1.2.2.1 Terminals 5 and 6. The contacts shall close with 0 volts across them. Approximately 1 second later an external circuit shall cause an inductive load at 30 vdc maximum, having a 112.5 ± 10 amp surge current (1.0 milliseconds, maximum, rise time) that lasts 0.3 second, minimum. At the end of the surge period, an external circuit shall switch the relay contacts to a 24 ± 1 amp inductive load at 30 vdc, maximum, using the same inductance as during the inrush. After 20 seconds the contact voltage is reduced to 0 volts.

4.4.2.10.1.2.2.2 Terminals 3 and 4. Zero current.

4.4.2.10.1.3 Group B - light duty contacts.

4.4.2.10.1.3.1 Terminals 7 and 8. Zero current.

4.4.2.10.1.3.2 Terminals 1 and 2. The contacts shall make and break a 12 ± 0.5 amp. dc resistive load at 30 vdc maximum.

4.4.2.10.1.4 Group B.

4.4.2.10.1.4.1 Heavy duty form A contacts only.

4.4.2.10.1.4.1.1 Terminals 5 and 6. Zero current.

4.4.2.10.1.4.1.2 Terminals 3 and 4. These terminals shall be subjected to the procedure specified in 4.4.2.10.1.2.1.1.

4.4.2.10.1.4.2 Heavy duty form B contacts only.

4.4.2.10.1.4.2.1 Terminals 5 and 6. Zero current.

4.4.2.10.1.4.2.2 Terminals 3 and 4. These terminals shall be subjected to the procedure specified in 4.4.2.10.1.2.2.1.

4.4.2.10.2 Applicable to classes 1 and 2 type III only. The relays shall be divided, designated and tested in accordance with 4.4.2.10.1, except that the coils shall be energized with 182 ma dc + 2 percent. The contact loads shall be as specified in 4.4.2.10.1.

4.4.2.11 Thermal cycle (see 3.7.9). The relay shall be subjected to three consecutive cycles of the following temperautre cycle:

Step	Air Temperature (°F.)	Time Interval
1	plus 160	4 hours
2	room ambient	5 minutes or less
3	minus 80	4 hours
4	room ambient	5 minutes or less

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All during Step 1 of the first cycle, the 220 ohms relay shall have applied to its coils, 30 vdc max, the 66 ohms relay shall have applied to its coil 17 volts max. At the end of Step 1 and while remaining in the test chamber, the coil temperature shall be determined by the resistive method. In no case shall this temperature exceed 260°F. During the last 3 hours of Step 1, of the third cycle, and while remaining in the test chamber, the relay shall be tested as specified in 4.4.1. Immediately after Step 3 of the third cycle, the relay shall be stabilized for four hours minimum at minus 65 + 5°F., and while remaining in the test chamber the relay shall be tested as specified in 4.4.1, except for 4.4.1.6 (dielectric strength). The relay shall then be stabilized at room ambient temperature and shall be tested as specified in 4.4.2.3.

4.4.2.12 Low temperature and pressure (see 3.7.10). The relay shall be subjected to a temperature of -65°F. for 3 hours, -80 + 5°F. for 48 hours minimum, and -65 + 5°F. for 24 hours minimum. During the final period of -65 + 5°F. exposure, the pressure shall be reduced to 2 inches Hg absolute. During the last 3 hours of the final -65 + 5°F. period, the relay shall be tested as specified in 4.4.1, except for 4.4.1.6 (dielectric strength). After the relay is stabilized at room ambient temperature, the relay shall be tested as specified in 4.4.2.3.

4.4.2.13 High pressure (see 3.7.11). The relay shall be placed in a chamber whose air pressure shall be increased to 45 inches Hg absolute, and maintained at this pressure while the relay is tested as specified in 4.4.1 and 4.4.2.5. At the conclusion of the test the relay shall be tested as specified in 4.4.2.3.

4.4.2.14 High temperature storage (see 3.7.12). The relay shall be subjected to a temperature of +160 + 5°F. with the relative humidity maintained at 15 percent or less, for 100 hours minimum. At the conclusion of this test, and after stabilization at room temperature for at least 3 hours, the relay shall be tested as specified in 4.4.1, 4.4.2.3 and 4.4.2.5.

4.4.2.15 Contact spring assembly. The manufacturer shall submit to the contracting officer samples of the contact spring assemblies for metallurgical examination in order to qualify their joining process.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The relays shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.1.2 Level C. The relays shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

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5.2 Packing.

5.2.1 Levels A and C. The relays shall be packed in accordance with the applicable instructions contained in MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

5.4 Data cards. Data cards shall be prepared in accordance with MIL-STD-1167.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 and 5.4 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.1.3 Type and corresponding drawing number.

Type I	-	8801686
Type II	-	8791622, 8790180, 8801687
Type III	-	8804004
Type IV	-	8803996
Type V	-	8790179

6.1.4 Contact circuit definitions.

Form A. A combination of a stationary contact and a movable contact which is unengaged when the coil is unenergized.

Form B. A combination of a stationary contact and a movable contact which is engaged when the coil is unenergized.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase description PA-PD-1166 (Rev. 2) dated 17 September 1963.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. NUOR-A043

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OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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